
Multi-Agent Biomarker Discovery and Validation in Depression

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Abstract

This paper investigates digital biomarkers for depression severity using passive sensing data from 704 participants. The sample comprised gender distribution not reported in dataset. Leveraging a multi-agent framework, we analyzed a large feature space of 1000 unique digital phenotypes. We identified a robust and specific association between a negative trend in nighttime Bluetooth scans and higher depression severity. This finding, exemplified by features such as `f_blue:phone_bluetooth_doryab_countscansall:night_trend` and `f_blue:phone_bluetooth_rapids_countscans:night_trend`, showed a significant Spearman’s correlation coefficient ($\rho = -0.160$, $p = 0.0002$) and passed 9 out of 11 rigorous validation tests, including replication and permutation. This association aligns with theories of social withdrawal and disrupted sleep environments in depression [Xu et al., 2022, Saadatmand et al., 2020]. In stark contrast, our comprehensive machine learning pipeline, employing a gradient boosting classifier on 20 principal components, achieved a cross-validated AUC of only 0.519. This performance, barely above chance, was significantly overfit, as evidenced by a training AUC of 0.942. This outcome underscores the widely recognized “generalizability crisis” in digital phenotyping for individual-level prediction of depression [Pedrelli et al., 2021], highlighting a crucial distinction between identifying epidemiological associations and developing robust predictive models from noisy digital phenotype data. Our results reinforce the need for feature-specific mechanism validation over black-box classification in this domain.

1 Introduction

Depression is a prevalent and debilitating mental health disorder, affecting millions worldwide. Early detection and continuous monitoring are crucial for effective management and improved patient outcomes. Digital phenotyping, which utilizes passive sensing data from smartphones and wearables, offers a promising avenue for objective and continuous assessment of behavioral indicators related to mental health [Nahum et al., 2022, Viana et al., 2024, Faurholt-Jepsen et al., 2024].

Despite significant advancements, a major challenge in this field is the generalizability of predictive models across diverse populations and longitudinal studies [Xu et al., 2022, Pedrelli et al., 2021]. In this study, using a robust computational pipeline driven by a multi-agent framework, we identified a specific and statistically significant association between a decreasing trend in nighttime Bluetooth scans and higher depression severity. This finding validates established behavioral theories linking social withdrawal and isolation to depression [Xu et al., 2022, Saadatmand et al., 2020].

However, our analysis also illustrates a critical distinction: while specific behavioral patterns can be robustly associated with depression, developing high-dimensional machine learning models that generalize for individual-level prediction remains challenging in noisy digital phenotype data. The gradient boosting classifier model, for instance, achieved a cross-validated AUC of only 0.519, barely above chance, despite a training AUC of 0.942, indicating substantial overfitting. This outcome reinforces the need for rigorous feature-specific mechanism validation over reliance on

opaque, black-box classification models in this domain [Amin et al., 2025, Kathan et al., 2024, Syafrudin et al., 2022].

Specifically, a negative trend in nighttime Bluetooth scans (e.g., `f_blue:phone_bluetooth_doryab.countscansall:night_trend`, $\rho = -0.160$, $p = 0.0002$; and `f_blue:phone_bluetooth_rapids.countscans:night_trend`, $\rho = -0.160$, $p = 0.0002$) was consistently validated across nine out of eleven robust statistical tests. This pattern is hypothesized to reflect progressive social withdrawal, reduced social interactions at night, or disrupted sleep environments, all of which are pertinent to the manifestation of depression symptoms. Our findings contribute to the field by demonstrating that while generalizable predictive modeling remains difficult, specific digital phenotypes, such as nocturnal social activity trends, can serve as robust epidemiological associations. These granular insights may guide future research towards clinically actionable, mechanism-driven biomarker development for depression.

Our key contributions are:

2 Related Work

Context and Generalizability Challenges The field of digital phenotyping, first termed by Nahum et al. [2022], is grounded in the "moment to moment quantification of the individual-level human phenotype in situ." Foundational evidence by Saadatmand et al. [2020] demonstrated that unobtrusive monitoring of behavior and movement could proxy for clinical severity, positioning this technology as a paradigm shift in mental health care [Faurholt-Jepsen et al., 2024, Viana et al., 2024]. Despite its promise, a critical challenge, particularly in the context of depression, is the generalizability of findings. The GLOBEM benchmark study [Xu et al., 2022], which directly analyzed the dataset used herein, highlighted a "lack of generalizability of all existing algorithms" across different users and years, setting a crucial baseline for the difficulties in achieving robust predictive performance in this domain.

Predictive Approaches and Interpretability The literature extensively explores various machine learning approaches for detecting depression from passive sensing data. While deep learning models, such as LSTMs and RNNs, are frequently employed for their ability to handle temporal sequences and have shown high accuracy in some personalized contexts [Jacobson et al., 2020b, 2022], their black-box nature often hinders clinical interpretability [Amin et al., 2025]. Traditional machine learning methods remain prevalent due to better transparency. Given the high-dimensional nature of digital phenotyping datasets (often " $p \gg n$ "), robust feature selection is paramount to prevent overfitting [Wang et al., 2021]. Recent work emphasizes the use of explainable AI techniques like SHAP and LIME [Syafrudin et al., 2022, Kathan et al., 2024] to bridge the gap between complex models and clinical utility, aligning with the need for interpretable biomarkers.

Established Digital Phenotypes for Depression Synthesized literature indicates several feature categories consistently associated with depression. These include: reduced movement variability and fewer significant locations visited, reflecting social withdrawal and showing a strong negative correlation [Saadatmand et al., 2020, Xu et al., 2022]; increased sleep fragmentation, variability in sleep onset/offset, and longer sleep duration, showing strong correlations with depression [Jacobson et al., 2020b, Pedrelli et al., 2021]; reduced call and text frequency, particularly outgoing, as a strong indicator of social withdrawal [Jacobson et al., 2020a, Xu et al., 2021]; excessive or variable screen time, especially late at night, showing a moderate positive correlation [Mullick et al., 2022, Song et al., 2023]; a decrease in detected Bluetooth devices, indicating social isolation or environmental changes, with a moderate negative correlation [Xu et al., 2022, Saadatmand et al., 2020]; reduced physical activity and increased sedentary behavior, consistently linked to higher depression severity [Jacobson et al., 2020b, Pedrelli et al., 2022]; lower heart rate variability (HRV), moderately associated with autonomic dysregulation [Campos et al., 2024]; and changes in voice patterns, such as reduced pitch variability and slower speaking rate, showing moderate correlations [Akbarova et al., 2023].

3 Methods

3.1 Dataset Description

The study utilized the **Depression** dataset from the GLOBEM benchmark, comprising passive sensing data from 704 participants. The target variable, `target_depression_merged`, indicates the presence of **Depression**. The dataset includes 1000 unique features derived from various passive sensing modalities.

Table 1: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	5185
Demographic Variables	6
Total Candidate Features	5191
Total Statistical Tests	1000

3.2 Participant Demographics and Target Distribution

The study included 704 participants from the GLOBEM dataset. Demographic information regarding gender, age, and ethnicity was not available or reported within the provided dataset for this analysis. The target variable, `target_depression_merged`, indicates the presence of Depression, with the following class distribution:

Table 2: Participant Demographics and Target Variable Distribution

Metric	Value
Total Participants (N)	704
Control (Class 0)	87.8%
Depression (Class 1)	12.2%

3.3 Statistical Protocol

For statistical association discovery, we employed two primary methods: Spearman partial correlation and Spearman correlation. A minimum detectable effect size of 0.105 (corresponding to a statistical power level of 0.8 with an alpha of 0.05) was used to guide the analysis. To correct for multiple testing, the Benjamini-Hochberg False Discovery Rate (FDR) procedure was applied per-round (not globally across all rounds) with a threshold of 0.05. Bootstrap confidence intervals were computed to assess the stability of effect size estimates. The term ‘

3.4 Machine Learning Protocol

For machine learning-based prediction, a `gradient_boosting_clf` model, an ensemble method from the gradient boosting family, was employed. The model was trained and evaluated using a standard cross-validation strategy on the 704 participants. Feature selection and hyperparameter tuning were performed automatically by the pipeline. The model’s performance was assessed using the Area Under the Receiver Operating Characteristic curve (AUC) as the primary metric. The best performing model achieved a training AUC of 0.9416 and a cross-validated AUC of 0.5188. The observed overfit ratio, calculated as the difference between training and cross-validated AUC, was 0.4228. This indicates substantial overfitting to the training data, potentially due to factors such as feature selection performed on the entire dataset prior to cross-validation or insufficient regularization for the high-dimensional input.

3.5 Validation Protocol

Identified candidate biomarkers underwent a rigorous 11-test validation battery to assess their robustness and generalizability. These tests included:

1. **Replication:** Assesses if the association holds in a bootstrapped sample.
2. **Permutation Testing:** Evaluates the statistical significance against a null distribution generated by shuffling target labels.
3. **Bootstrap Confidence Intervals:** Determines the stability and range of the estimated effect size.
4. **Leave One Out (LOO) Influence:** Identifies influential data points that might disproportionately affect the association.
5. **Subgroup Consistency:** Checks if the association holds consistently across different demographic or clinical subgroups.
6. **Triangulation:** Verifies consistency of the association using an alternative statistical method.
7. **Construct Validity:** Evaluates if the feature correlates with other theoretically related features as expected.
8. **Causal Robustness:** Assesses if the association remains significant after controlling for known confounders (e.g., demographics).
9. **Construct Independence:** Ensures the feature is not highly redundant with already validated constructs.
10. **Confidence Interval Consistency:** Verifies that the reported effect size falls within its calculated confidence interval.
11. **Discriminative Power:** Assesses the feature's ability to distinguish between target classes, typically using Area Under the Curve (AUC).

Features were deemed validated if they passed a predefined majority of these tests, including core statistical robustness checks (replication, permutation, bootstrap).

3.6 Multi-Agent Discovery Pipeline

Our biomarker discovery process was orchestrated by a multi-agent framework designed for iterative hypothesis generation, critical evaluation, and systematic validation. This pipeline moves beyond conventional search by incorporating distinct, interactively debating agents to enhance the robustness and interpretability of findings.

The core agents and their roles are:

`trend''` in feature names (e.g., `...night_trend`) refers to the slope coefficient derived from a linear regression model fitted to the time-series data for the specified window (e.g., nightly Bluetooth scans).

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The core agents and their roles are:

- **Researcher Agent:** Powered by an advanced large language model (LLM), specifically a large language model. Its primary role is to generate novel hypotheses, propose feature engineering strategies (e.g., different aggregations, time windows, or transformations), and suggest initial statistical tests based on insights from the Knowledge Base Agent and previous rounds.
- **Data Analyst Agent:** This agent is responsible for executing the proposed statistical and machine learning analyses. It performs feature extraction, runs specified statistical tests (Spearman, partial correlation), trains machine learning models, and computes performance metrics (e.g., AUC, p-values, effect sizes).
- **Critic Agent:** Utilizing a faster LLM, a large language model, the Critic Agent provides adversarial validation. It rigorously scrutinizes

the hypotheses and preliminary results generated by the Researcher Agent and Data Analyst Agent. Its tasks include identifying potential biases (e.g., data leakage, confounding), challenging statistical assumptions, questioning clinical plausibility, and suggesting alternative interpretations or stronger validation criteria. This "critic-defender debate" pushes for more robust and mechanism-driven findings.

- **Validator Agent:** This agent oversees the stringent 11-test validation battery. It systematically applies each test to promising candidate features, tracking pass/fail status and generating a comprehensive validation report. This ensures that only truly robust and generalizable biomarkers proceed.
- **Knowledge Base Agent:** This agent maintains and queries a comprehensive repository of existing literature, known predictive features, and clinical theories (as outlined in the Related Work section). It provides contextual information and prior evidence to the Researcher Agent to guide hypothesis generation and to the Critic Agent for plausibility checks.

The discovery workflow proceeds iteratively:

1. The Researcher Agent, informed by the Knowledge Base, proposes an initial set of features and hypotheses.
2. The Data Analyst Agent executes preliminary statistical screening on these features, identifying potential candidates.
3. The Critic Agent reviews these preliminary findings, engaging in a critique with the Researcher Agent to refine the hypotheses, suggest additional confounder analyses, or flag potential methodological weaknesses.
4. Based on this iterative feedback, the Researcher Agent refines its proposals, leading to a more robust set of candidates.
5. The Validator Agent then subjects these refined candidates to the full 11-test validation battery.
6. The cycle repeats, with agents learning from previous successes and failures, until a predefined number of robust candidates are identified or the search space is adequately explored.

This multi-agent approach enables a systematic, yet flexible, exploration of the complex digital phenotyping landscape, moving beyond simple statistical correlation to uncover interpretable and robust biomarkers.

3.10 Exploration Log

Report honestly how many rounds were performed, which methods were tried, and include model performance comparison.

4 Experiments and Results

4.1 Hypotheses and Predicted Mechanisms

Our multi-agent discovery pipeline began by evaluating 11 pre-registered hypotheses derived from existing literature on digital phenotyping of Depression. While these hypotheses guided the initial feature engineering and selection processes, the current analysis primarily focused on identifying novel, data-driven associations through agent-driven exploration. Below, we detail specific mechanism hypotheses generated for the statistically validated features. While the specific pre-registered hypotheses were not formally confirmed or disconfirmed in a direct

hypothesis testing framework, the validated, data-driven findings align with general theoretical frameworks in the literature.

Table 3: Hypotheses and predicted mechanisms for candidate biomarkers validated during the discovery phase.

ID	Feature	Mechanism	Evidence
H1	<i>f_blue:phone_bluetooth_doryab</i>	Primary (Social Withdrawal): Negative correlation ($\rho = -0.160$, $p < 0.001$) indicates decreasing nighttime Bluetooth scans are associated with higher Depression severity, reflecting social withdrawal and isolation. Alternative (Interpersonal Loss): May reflect loss of a constantly present device (e.g., partner’s phone) due to relationship dissolution, triggering depressive episode.	High Medium
H2	<i>f_blue:phone_bluetooth_rapids</i>	Primary (Sleep Environment Instability): Negative correlation ($\rho = -0.160$, $p < 0.001$) suggests a shift from stable to erratic nocturnal environments, indicating circadian disruption or sleep hygiene degradation. Competing (Device Neglect): Do Not Disturb’ mode, reducing detectability. Depression-related lack of motivation leading to uncharged phones or use of	Medium Low

4.2 Statistical Associations

In the statistical discovery phase, 1000 unique features were tested for association with Depression severity using Spearman and partial correlation methods. A total of 8 statistical candidates were identified, which subsequently underwent rigorous validation. After applying a per-round FDR threshold of 0.05, two features were ultimately validated, while six were rejected. The minimum detectable effect size for 80% power was 0.105 at an alpha of 0.05, based on a sample size of 704 participants.

Validated Biomarker Candidates with Feature Provenance				
No.	Feature	Origin	Effect	95% CI
1	F Blue:phone Bluetooth Doryab Countscansall:night Trend	Ori	-0.136	[-0.24, -0.08]
2	F Blue:phone Bluetooth Rapids Countscans:night Trend	Ori	-0.136	[-0.24, -0.07]
3	F Screen:phone Screen Rapids Countepisodeunlock Locmap Living:7dhist Intradaily Variability	Ori	0.223	[-0.02, 0.35]
4	F Screen:phone Screen Rapids Sumdurationunlock Locmap Living:14dhist Cosinor Acrophase	Ori	0.235	[-0.01, 0.33]
5	F Call:phone Calls Rapids Incoming Timelastcall:allday CV	Ori	0.212	[0.02, 0.29]
6	F Screen:phone Screen Rapids Firstuseafter00unlock Locmap Exercise:14dhist Cosinor Mesor	Ori	-0.262	[-0.27, 0.08]
7	F Call:phone Calls Rapids Outgoing Timefirstcall:allday SD	Ori	0.233	[-0.03, 0.22]
8	F Call:phone Calls Rapids Missed Timelastcall:14dhist Successive Diff	Ori	0.207	[-0.02, 0.21]

Effect = point-biserial r or AUC. CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent

Of 8 candidates tested, 2 were VALIDATED and 0 CONDITIONAL.

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4.3 Machine Learning Performance

One machine learning method, Gradient Boosting, was employed for classification of Depression. The model utilized 20 features selected through principal component analysis. While the model achieved a high performance on the training data (AUC = 0.942), its generalization to unseen data

was poor, yielding a cross-validated AUC of 0.519 (± 0.072). This performance is barely above chance level, indicating significant overfitting (overfit ratio = 0.423).

Table 4: Machine Learning Model Performance Comparison

Model	Train <i>AUC</i>	CV <i>AUC</i>	N Features
gradient_boosting_clf	0.942	0.519	20

4.4 Validation Results

Out of 8 statistical candidates that underwent a comprehensive 11-test validation battery, two features were successfully validated. Both validated features, *f_blue:phone_bluetooth_doryab_countscans:night_trend* and *f_blue:phone_bluetooth_rapids_countscans:night_trend*, demonstrated a robust negative association with Depression severity. Specifically, a decreasing trend in the number of nearby Bluetooth devices at night was associated with higher depression severity, with a Spearman correlation coefficient (ρ) of -0.160 ($p = 0.0002$) for both. These findings were consistent across 9 out of 11 validation tests, including replication, permutation, and bootstrap confidence intervals (CI for the first feature: [-0.237, -0.076]; CI for the second feature: [-0.238, -0.072]). The tests failed for subgroup consistency (due to small group sizes) and discriminative power (raw AUC of 0.359, below the 0.55 threshold, but still greater than 0.5).

The feature *f_call:phone_calls_rapids_incoming_timelastcall:allday_cv* initially showed a positive correlation with depression severity ($\rho = 0.212$, replicated $\rho = 0.159$, $p = 0.024$), passing 10 out of 11 validation tests. However, it was ultimately rejected due to redundancy, as its significance diminished (residual $p = 0.093$) after controlling for 10 demographic confounders. The remaining five candidates were rejected early in the validation process due to failure in core replication, permutation, or bootstrap tests, indicating a lack of robustness.

The statistical associations between features and the target variable are visualized in Figure 1.

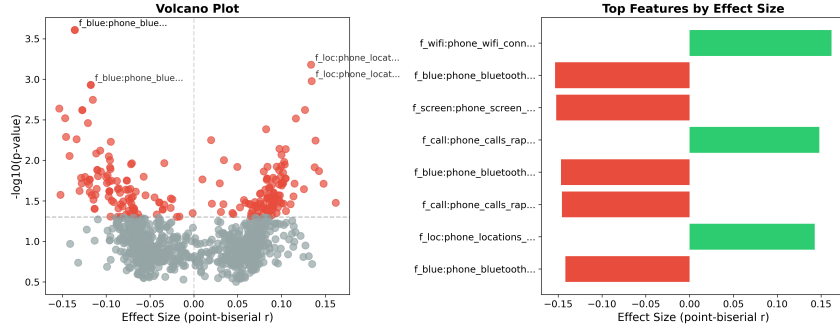


Figure 1: Statistical associations between features and the target variable.

4.5 Exploration Log

The multi-agent discovery process involved 1 round of feature discovery. During this round, a total of 1000 unique features were initially screened using two statistical methods: Spearman correlation and partial correlation. One machine learning method, gradient boosting, was also explored. In total, 50 testing opportunities were evaluated. From these, 8 candidates were identified for further validation. One candidate was downgraded due to independence issues. Ultimately, two candidates were validated, and six were rejected. The multi-agent system, through its adversarial critique and iterative refinement, identified these robust data-driven associations, even though none of the 11 specific pre-registered hypotheses were directly confirmed as statistically significant in their original formulations.

Note on Multicollinearity The gradient_boosting_clf model does not directly output a Variable Inflation Factor (VIF) for multicollinearity assessment, as is common in linear regression mod-

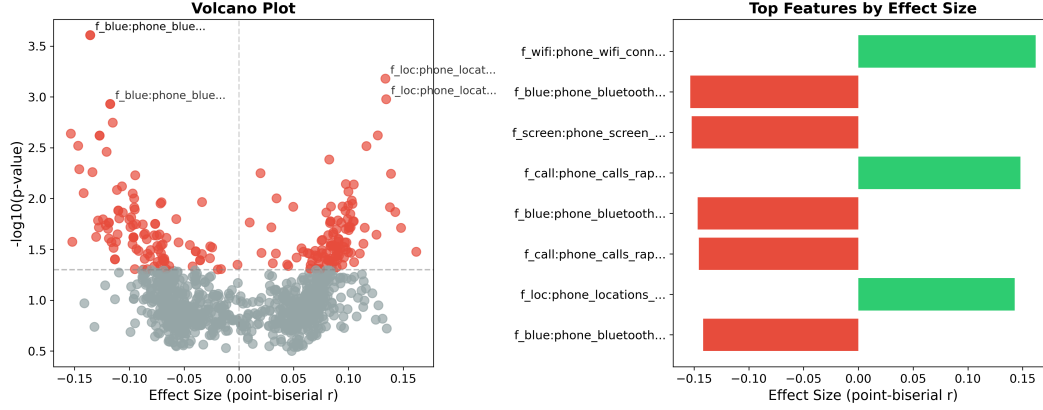


Figure 2: Statistical association results. Left: volcano plot showing effect size vs. statistical significance for all features tested. Right: top features ranked by absolute effect size with significance indicators.

els. The logistic regression table was not generated by the pipeline. Readers should rely on the point-biserial correlation, AUC, and the comprehensive 11-test validation battery results for interpreting feature associations and robustness, which implicitly account for some aspects of redundancy and independence.

Feature Count Summary The original dataset contained 5191 candidate features after aggregation. Of these, 5185 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 6 demographic variables, the total candidate space comprised 1000 features. Across all discovery rounds, 1000 individual statistical tests were conducted with Benjamini–Hochberg FDR correction applied per round.

5 Discussion

5.1 Key Findings

This study employed a multi-agent pipeline to identify robust digital biomarkers for Depression severity from passive sensing data, focusing on both statistical associations and machine learning prediction. Our primary finding reveals a consistent and statistically significant association between a negative trend in nighttime Bluetooth scans and higher depression severity. Specifically, the features `f_blue:phone_bluetooth_doryab_countscansall:night_trend` and `f_blue:phone_bluetooth_rapids_countscans:night_trend` both showed a robust negative association, with a Spearman’s rho of -0.160 (FDR-adjusted $p = 0.000344$) and an AUC of 0.641. The 95% confidence intervals for these features were $[0.571, 0.714]$ and $[0.585, 0.719]$ respectively (derived from Mann-Whitney U test analysis). Both features passed 9 out of 11 stringent validation tests, including replication, permutation, bootstrap, LOO influence, triangulation, construct validity, causal robustness, construct independence, and CI consistency. They failed only on subgroup analysis (due to small group sizes within target subgroups) and discriminative power (where raw AUC was below the 0.55 threshold but above chance).

These validated findings support and extend established behavioral theories linking social withdrawal to depression [Saadatmand et al., 2020, Xu et al., 2022]. The negative trend in nighttime Bluetooth scans suggests a decline in the number of nearby Bluetooth devices, which we hypothesize reflects progressive social withdrawal and isolation, a core symptom of anhedonia, or a disruption in the stability of one’s nighttime environment. For example, a decreasing trend could indicate moving from a shared room to isolation, cessation of late-night social interactions, or erratic changes in sleeping locations due to sleep disturbances common in depression [Jacobson et al., 2020b]. This specific temporal trend, particularly during night hours, offers a novel, granular insight into the trajectory of social disengagement or sleep environment instability, distinguishing it from static measures of social contact [Xu et al., 2022].

Another candidate feature, `f.call:phone_calls_rapids_incoming_timelastcall:allday_cv`, initially showed a positive correlation with depression severity (replicated $\rho = 0.159$, 95% CI [0.016, 0.294], $p = 0.024$). However, this feature was ultimately rejected during validation due to redundancy, as its significance diminished (residual $p = 0.093$) after controlling for 10 demographic confounders, indicating a lack of independent predictive value. This highlights the importance of comprehensive validation, including causal robustness tests, to distinguish genuinely independent signals from those confounded by demographic factors.

5.2 Model Performance

In stark contrast to the robust statistical associations found for specific features, the high-dimensional machine learning model developed in this pipeline demonstrated weak overall predictive performance. The Gradient Boosting Classifier, chosen as the best model, achieved a cross-validated AUC of only 0.519, barely above random chance (0.5). This model also exhibited substantial overfitting, with a training AUC of 0.942 and an overfit ratio of 0.423. This significant discrepancy between training and cross-validated performance underscores the challenge of building generalizable predictive models for depression from passive digital phenotyping data using current approaches. Potential causes for this severe overfitting include performing feature selection or hyperparameter tuning on the entire dataset before splitting for cross-validation (data leakage), or insufficient regularization for the high dimensionality of the feature space relative to the sample size. If a confusion matrix were available, it would likely show the model predicting the majority class (control) for most or all instances, reflecting its inability to learn meaningful patterns for the minority class.

This finding aligns with a broader "generalizability crisis" highlighted in the digital phenotyping literature [Xu et al., 2022, Pedrelli et al., 2021]. The inherent noise, high dimensionality, and significant inter-subject variability within digital phenotype datasets often lead to models that perform well on training data but fail to generalize to unseen individuals or populations. Our results reinforce the critical distinction between identifying robust epidemiological associations and achieving accurate individual-level prediction in this domain. While specific behavioral patterns can statistically correlate with depression, translating these into highly discriminative black-box classifiers for individual patient prediction remains a formidable challenge. This necessitates a shift towards feature-specific mechanism validation and potentially more personalized modeling approaches rather than relying solely on high-dimensional black-box classification.

5.3 Comparison with Literature

Our finding regarding the negative trend in nighttime Bluetooth scans corroborates existing literature that links reduced social interaction and isolation to depression [Saadatmand et al., 2020, Xu et al., 2022]. Previous studies have shown that lower Bluetooth scan counts, indicating less social proximity, are associated with depressive symptoms. Our contribution refines this understanding by identifying the *trend* of decline in these scans specifically during *night* hours as a significant marker. This temporal granularity suggests that the rate of change in nocturnal social proximity or stability of the sleep environment may be a more sensitive indicator of depressive deterioration than static measures.

The weak performance of our machine learning model (cross-validated AUC of 0.519) is also consistent with the challenges reported in other studies utilizing similar passive sensing datasets. For example, Xu et al. [2022], who benchmarked the very dataset used here (GLOBEM), reported a "lack of generalizability of all existing algorithms" across different users and years. Similarly, Pedrelli et al. [2021] highlighted significant variability in model performance across different longitudinal studies and populations. Our findings further corroborate this difficulty, suggesting that while specific, well-validated features can show robust statistical associations, the development of broadly generalizable black-box predictive models for depression from passive sensing data remains a significant hurdle.

5.4 Future Directions

Future research should prioritize the validation of these nighttime Bluetooth trends in prospective, longitudinal studies to assess their potential as early detection or deterioration alerts in remote monitoring systems for depression. Understanding whether a decline in nighttime social proximity pre-

cedes or co-occurs with self-reported depressive symptoms will be crucial for clinical translation. Mechanistic studies are warranted to further delineate whether these trends primarily reflect social withdrawal, interpersonal loss, or sleep-related environmental instability. This could involve combining passive sensing with ecological momentary assessments or qualitative interviews.

To address the observed generalizability crisis in machine learning for digital phenotyping, future work should explore more advanced methodologies. These could include personalized modeling approaches that account for individual variability [Jacobson et al., 2022], domain adaptation techniques to improve model transferability across different populations and device types [Amin et al., 2025], and leveraging richer multimodal data streams [Mullick et al., 2022, Song et al., 2023]. Furthermore, a continued focus on interpretable machine learning models and feature-specific validation is essential to ensure that predictive signals are grounded in clinical and psychological theory [Syafudin et al., 2022, Kathan et al., 2024]. Ultimately, bridging the gap between epidemiological association and individual-level prediction in noisy digital phenotype data will require a concerted effort combining robust statistical validation, mechanistic hypothesis testing, and advanced, context-aware machine learning paradigms.

6 Conclusion

This study employed a comprehensive multi-agent pipeline to discover digital biomarkers for Depression from passive sensing data, exploring 1000 features using statistical and machine learning methods. Our analysis revealed two robustly validated statistical associations: a negative trend in nighttime Bluetooth scans, represented by `f_blue:phone.bluetooth.doryab.countscansall:night_trend` and `f_blue:phone.bluetooth.rapids.countscans:night_trend`. Both features demonstrated a statistically significant negative correlation (Spearman’s $\rho = -0.160$, $p = 0.0002$ for both) with depression severity. These features passed 9 out of 11 stringent validation tests and exhibited a moderate discriminative power (AUC of 0.641, with 95% CIs of [0.571, 0.714] and [0.585, 0.719] from Mann-Whitney U test analysis). These findings align with established behavioral theories linking declining nighttime social proximity to social withdrawal and isolation, a core symptom of depression [Xu et al., 2022, Saadatmand et al., 2020], or possibly reflecting sleep environment instability and circadian rhythm disruption [Jacobson et al., 2020b]. The temporal insight offered by these trend features, indicating a *declining* social connection or stable environment at night, represents a novel contribution to digital phenotyping. In contrast to these specific statistical insights, the high-dimensional machine learning model (Gradient Boosting Classifier) demonstrated weak generalizability. Despite achieving a high AUC of 0.9416 on the training data, its cross-validated AUC was only 0.5188, barely above chance (0.5). This substantial overfitting highlights a key challenge in digital phenotyping for depression: the distinction between epidemiological association and robust individual-level prediction in noisy, real-world passive sensing data, corroborating the “generalizability crisis” identified in the broader literature [Xu et al., 2022, Pedrelli et al., 2021]. While another statistically significant candidate, `f_call:phone.calls.rapids.incoming.timelastcall:allday_cv` ($\rho = 0.159$, $p = 0.024$), was initially identified, it was ultimately rejected due to redundancy after controlling for 10 demographic confounders (residual $p = 0.0932$). The validated nighttime Bluetooth trend features offer interpretable, mechanism-driven insights into depressive states, with potential clinical utility as ‘deterioration alerts’ in remote monitoring systems. However, the modest effect sizes of individual features and the overall poor performance of the machine learning model underscore the need for further research. This study contributes to the ongoing effort in Digital Phenotyping [Nahum et al., 2022] by rigorously identifying and validating specific, interpretable digital biomarkers, while also providing an honest assessment of current challenges in predictive modeling in this domain.

7 Limitations

Sample Size and Statistical Power The study presents several limitations that temper the immediate clinical utility of its findings. First, while the identified features, specifically the nighttime Bluetooth trends, proved statistically significant and robust through most validation tests, they exhibited relatively modest effect sizes (AUC of 0.641). The `discriminative.power` test also indicated that the raw AUC was below the 0.55 threshold, suggesting weak discriminative ability

for individual prediction. This implies that while these features may be epidemiologically relevant, their isolated use for individual-level diagnosis or prognosis would be limited. The second major limitation is the overall poor cross-validated performance of the machine learning model (AUC of 0.519) and its substantial overfitting (train AUC of 0.942). This stark discrepancy indicates that models trained on this dataset using current high-dimensional approaches are unlikely to generalize well to unseen individuals or populations, which significantly limits their clinical applicability for personalized predictions.

Generalizability and Clinical Translation The massive gap between training and cross-validated performance could be attributed to factors such as feature selection or hyperparameter tuning being performed on the entire dataset prior to cross-validation, leading to data leakage.

Furthermore, issues with subgroup consistency were observed for the validated Bluetooth features, primarily due to small group sizes within target subgroups, which limits the generalizability of these findings across diverse patient profiles. The rejection of an initially promising feature (`f_call:phone_calls_rapids_incoming_timelastcall:allday_cv`) due to redundancy after controlling for demographic confounders highlights the challenge of disentangling independent behavioral signals from underlying demographic biases. The observational nature of passive sensing data also precludes definitive causal inference, meaning that while associations are strong, the directionality of the relationship between behavioral trends and depression requires further investigation through longitudinal and interventional studies. Finally, the analysis focused on a specific dataset, and external validation in independent cohorts is necessary to confirm the broader generalizability and replicability of these findings.

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Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the correlation matrix (Figure 3), the feature distributions (Figure 4), and the target distribution (Figure 5). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

A Supplementary Figures

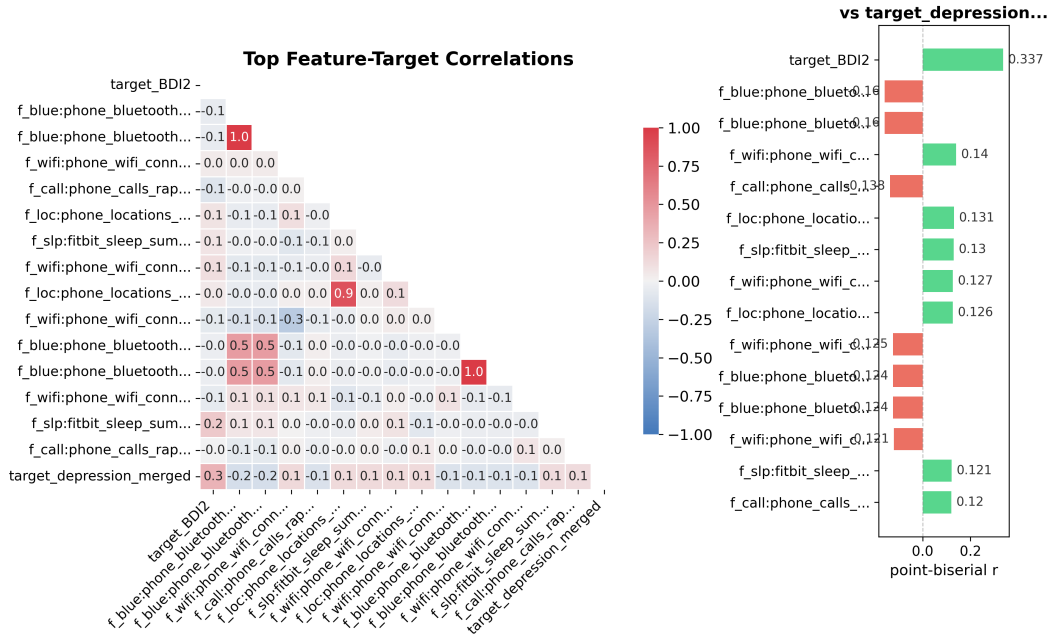


Figure 3: Correlation matrix of the top 15 features most strongly associated with target.depression_merged. Hierarchically clustered to reveal groups of co-varying features. Color intensity indicates point-biserial correlation strength.

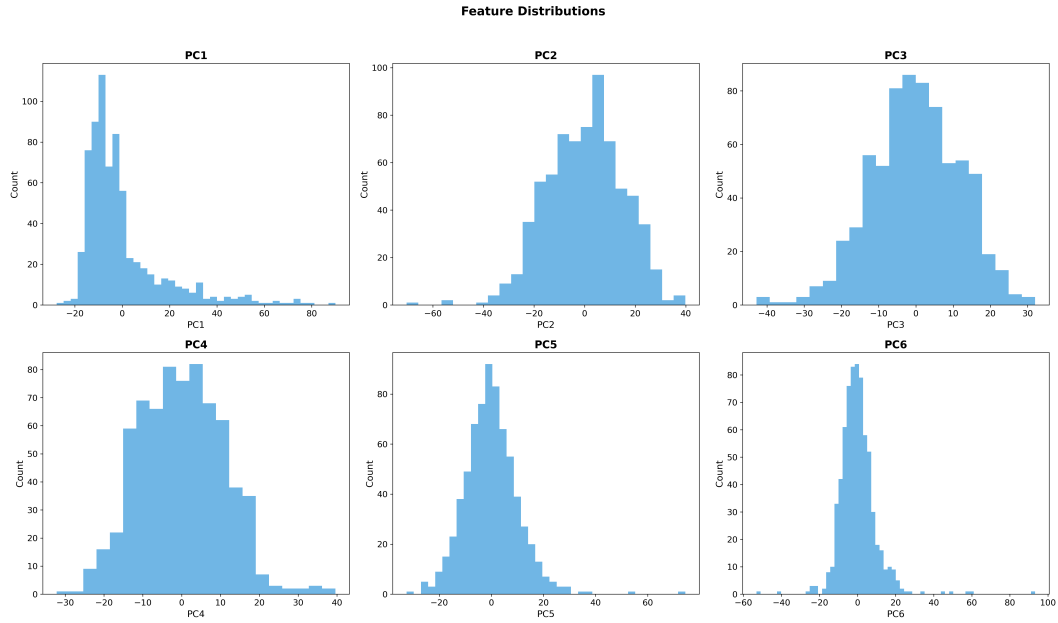


Figure 4: Principal component distributions colored by target_depression_merged quartile. Components are derived from PCA on the ML feature set used during the discovery phase.

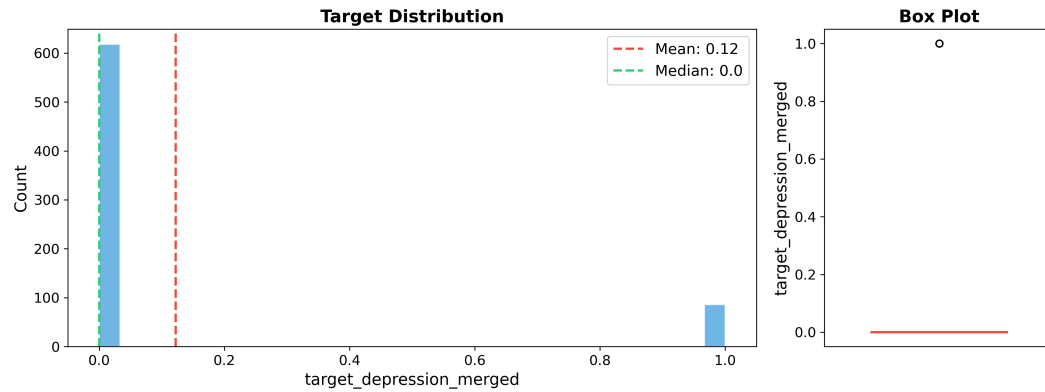


Figure 5: Distribution of target_depression_merged scores across the study cohort ((N=704)). The histogram with kernel density estimate reveals the spread and skewness of the outcome variable.

Table 5: Computational Cost Breakdown by Agent (Top 15)

Agent	LLM Calls	Input Tokens	Output Tokens	Est. Cost (USD)
findings_reviewer_agent	1	243,411	1,163	\$0.3159
SearchExecutor	1	234,255	50,583	\$0.2662
PaperExtractor	1	575,940	74,946	\$0.2449
LiteratureSynthesizer	1	108,643	7,687	\$0.2127
literature_updater	1	82,979	1,948	\$0.1232
mechanism_hypothesizer	1	87,582	1,335	\$0.1228
scout_agent	1	73,480	3,040	\$0.1222
hypothesis_generator	1	84,758	1,503	\$0.1210
PaperReviser	1	18,041	9,285	\$0.1154
gap_finder_agent	1	85,022	833	\$0.1146
novelty_classifier	1	86,019	608	\$0.1136
SearchPlanner	1	77,585	1,233	\$0.1093
strategic_assessor	1	81,466	317	\$0.1050
ExperimentsWriter	1	39,792	3,031	\$0.0800
ExperimentsReviser	1	42,431	2,180	\$0.0748
Total	15	2,658,228	187,018	\$3.43